NATIONAL UNIVERSITY

BANGLADESH



Syllabus
Department of Psychology

Four Year B.Sc (Honours) Course
Effective from the Session: 2024–2025

Mission

To promote "Progress for Everyone" by fostering transformative education, conducting pioneering research, and encouraging meaningful collaboration with individuals, communities, and partners both in Bangladesh and around the world.

Vision

We expect that the hundredth anniversary of the National University, our commitment to "Progress for Everyone" will be evident in all areas of our operations and stakeholder engagements.
We leverage our core strength in education, research, and community engagement to address pressing global issues and foster sustainable development.
By 2042, National University aims to strengthen its positions as Bangladesh's top academic institution and rank among the world's leading universities. Our focus will be on expanding expertise building strategic partnerships, and increasing our global influence. Our services will be measured by the achievements of the individuals and communities we serve.

Name of the Program: B.Sc. (Honours) in Psychology

Program Educational Objectives (PEOs)

PEO 1: Technical and Professional Competence

Graduates will demonstrate strong theoretical knowledge and practical skills in the field of Psychology enabling them to meet contemporary market demands in education, research, planning, and development sectors.

PEO 2: Research and Problem Solving

Graduates will be equipped to carry out collaborative, multidisciplinary research, evaluate environmental problems critically, and offer creative, evidence-based solutions to promote sustainable development in Bangladesh and elsewhere.

PEO 3: Social Responsibility and Ethics

Graduates will demonstrate a strong sense of social, ethical, and environmental responsibility, making valuable contributions to environmental management, community development, and the accomplishment of the Sustainable Development Goals (SDGs).

PEO 4: Lifelong Learning and Leadership

Graduates will pursue lifelong learning through continuous education and professional development, demonstrating leadership, teamwork, and communication skills essential for adapting to the dynamic global landscape.

Program Learning Outcomes (PLOs):

At the time of graduation, B.Sc. (Honours) in Psychology students will have achieved the ability to:

PLO 1: Core Disciplinary Knowledge: Demonstrate a comprehensive understanding of fundamental concepts, theories, and frameworks in cognitive, developmental, social, clinical, biological, and educational psychology.

PLO 2: Application of Knowledge: Apply psychological theories and evidence-based practices to understand and address individual, group, and societal issues across diverse contexts.

PLO 3: Problem Solving and Critical Thinking: Identify, analyze, and propose solutions to psychological and behavioral problems using scientific reasoning, critical thinking, and ethical decision-making.

PLO 4: Research and Innovation: Design and conduct psychological research using appropriate methodologies for data collection, analysis, and interpretation, contributing to innovative insights into human behavior and mental processes.

PLO 5: Use of Modern Tools: Effectively utilize modern tools and technologies in psychological assessment, data analysis (e.g., SPSS, R), virtual labs, and digital mental health platforms, adhering to scientific and ethical standards**.**

PLO 6: Communication Skills: Communicate psychological concepts, data, and findings clearly and effectively in oral, written, and visual forms to both specialist and non-specialist audiences.

PLO 7: Teamwork and Leadership: Demonstrate collaborative and leadership abilities in diverse teams, particularly in therapeutic, educational, research, and organizational settings**.**

PLO 8: Professionalism and Ethics: Exhibit professional integrity, ethical conduct, empathy, and social responsibility in psychological practice, research, and community engagement.

PLO 9: Lifelong Learning: Recognize the importance of continuous learning and professional development to stay updated with evolving psychological practices, societal needs, and scientific advancements.

PLO 10: Societal and Environmental Responsiveness: Address psychological needs and challenges in local and global contexts, with sensitivity to cultural, societal, and contextual factors, aligning practices with the Sustainable Development Goals (SDGs).

PLO 11: Entrepreneurial and Employment Readiness: Demonstrate the skills, attitudes, and knowledge necessary for employment in clinical, counseling, academic, research, corporate,

developmental, and non-governmental sectors, including entrepreneurial ventures in mental health and well-being.

PLO 12: Project and Time Management: Effectively plan, organize, and execute academic and field-based psychological projects, demonstrating strong time management, responsibility, and independent initiative.

Mapping PEO with PLO:

Program Learning Outcomes (PLOs)	PEO 1	PEO 2	PEO 3	PEO 4
PLO-1	✓			
PLO-2	✓	✓		
PLO-3	✓	✓	✓	
PLO-4	✓	✓		✓
PLO-5				
PLO-6	✓		✓	✓
PLO-7	✓		✓	✓
PLO-8	✓		✓	✓
PLO-9				✓
PLO-10		✓		✓
PLO-11	✓			✓
PLO-12	✓	✓		✓

Courses and Marks Distribution

First Year

Course Code	Course Title	Marks	Credits
213401	Introduction to Psychology	100	4
213403	Experimental Psychology	100	4
213405	Statistics in Psychology I	100	4
213406	Practical: Introduction to Psychology	100	4
219901	History of Bangladesh: Language, Culture & Identity	100	4
219903	Information and Communication Technology	75	3
219904	Lab: Information and Communication Technology	25	1
Any Three Subjects from the following			
212009	Introducing Sociology **OR**	100×3 =300	3×4 =12
212109	Introduction to Social Work		
211909	Introduction to Political Science		
211709	Introduction to Philosophy		
213207	Introduction to Geography and Environment		
213107	Zoology –I		
	Total =	900	36

Course Code:	213401	Marks: 100	Credits: 4	Class Hours:60
Course Title:	Introduction to Psychology			

Course Description

This course provides an introduction to psychology, covering the various approaches, subfields, and methods in the discipline. Students will explore the biological foundations of behavior, sensation and perception, learning, memory, language, and problem-solving. The course also delves into theories of motivation, emotion, personality, and intelligence, providing a broad understanding of psychological concepts and processes. It will offer a foundation for understanding human behavior from both a scientific and practical perspective.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	Define the key concepts of psychology, including its nature as a science, and explain the different approaches to the study of psychology (neurobiological, behavioral, cognitive, psychoanalytic, and humanistic approaches)
CLO2	Analyze the biological bases of behavior, including the structure and function of the nervous system, neural transmission, and the major divisions of the nervous system
CLO3	Apply of sensation and perception to real-world scenarios, understanding sensory thresholds, attention, and depth perception
CLO4	Demonstrate an understanding of the learning process, including classical and operant conditioning, and apply these theories to human behavior and cognitive development

Mapping CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	√	√										√
CLO2	√						√	√				
CLO3		√		√	√							
CLO4			√	√						√	√	

Course Contents

Chapter	Topics	Teaching-Learning Strategies	CLOs
1	**Introduction to Psychology:** Definition and nature of psychology, Psychology as a science, Approaches to the study of psychology: neurobiological approach, behavioral approach, cognitive approach, psychoanalytic approach, humanistic approach, The subfields of psychology: experimental and physiological psychology, clinical and counseling psychology, industrial, personnel and engineering psychology, school and educational psychology, social psychology, developmental psychology, personality psychology; The methods of psychology: experimental method, observational method, archival research, case history, survey method	Lectures, handouts, power point presentations, seminar presentation	CLO1
2	**Biological Bases of Behavior:** Cells, tissues, organs and systems of the body, Basic units of nervous system: neurons, structure and connectors , Transmission of neural impulses, The major divisions of nervous system: the central nervous system, brain and spinal cord, the peripheral nervous system-somatic and autonomic system, Glands: exocrine and endocrine glands; genetic influences of behavior	Multimedia lecture, group activity on cosmic timeline, guided Q&A	CLO1

3	**Sensation and Perception:** What is sensation? Measuring sensory experience- absolute, differential, and terminal thresholds, Visual auditory and other senses, The nature of perception, Selectivity in perception: selective attention, determiners of stimulus selection, Form perception: figure and ground, contour perceptual organization, Perceptual constancy-size and brightness constancy, Depth perception- monocular and binocular cues to depth perception	PPT-based lecture, discussion, solar system model interpretation, visual quiz	CLO2
4	**Learning:** Definition of learning, Classical conditioning: processes in classical conditioning, classical conditioning and human behavior, Operant conditioning: processes on operant conditioning, types of reinforcement, schedules of reinforcement, shaping, Cognitive mapping: latent learning, insight learning, observational learning	Diagram demonstration, 3D globe activity, interactive lecture, problem-solving	CLO2
5	**Memory and Forgetting:** Defining memory, Types of memory: sensory memory, short-term memory, long term memory, Memory dysfunction, Improving memory, The physiology of memory, Why do we forget? interferences, decay, retrieval failure, repression	Lesson plans, class lectures, handouts, PowerPoint presentations, seminar presentations	CLO2
6	**Language, Thinking and Problem Solving:** The structure and rules of language, How language is learned, Language and concepts, The thinking process, Problem solving	Lecture with animation, diagram-based demonstration, almanac analysis, class discussion	CLO3
7	**Motivation:** Defining motivation, Theories of motivation: instinct theory, drive theory, arousal theory, opponent process theory, incentive theory, cognitive theory, Maslow's Hierarchy Theory of motivation, Classification of motives: biological motives, social motives, Activity and stimulation needs, Motivation and behavior	Timeline construction, comparative charting, calendar workshop, case-based learning	CLO3
8	**Emotion:** Defining emotion, Theories of emotion: James-Lange Theory of emotion, Cannon-Bard Theory of emotion and Schachter-Singer Theory of emotion, Physiological correlates of emotion, The brain and emotions, Primary emotions	Case studies (real-life emotional responses), video-based observation and analysis of emotions, role-play to demonstrate theories,	CLO3

		group discussion on emotional experiences, lecture with multimedia animation	
9	**Personality:** Defining personality, Determinants of personality, Approaches to personality: Freud and Neo-Freudians, Allport-identifying basic characteristics, Cattle & Eysenck: factoring out personality, Are we born with personality? Measuring personality	Interactive lectures, personality test demonstrations, group discussions, case study analysis, self-reflection exercises	CLO4
10	**Intelligence and Creativity:** Definition of intelligence, Measurement of intelligence, Representative intelligence tests, Creativity and intelligence	Demonstration of IQ tests, group problem-solving tasks, creative thinking workshops, case studies on giftedness, interactive lectures, brainstorming sessions	CLO4

List of Books

Crider, A.B., Goethals, G.R., Roberts, D. Kavanaugh, R.D., Solomon, P.R. (1993). Psychology. New York: Harper Collins college publishers.

Coon. D. and Martini, T. S. (2018). Introduction to Psychology: Gateways to Mind and Behavior. Cengage Learning.

Feldman, R. S. (2019). Understanding Psychology. New York: McGraw Hill Education.

Morgan, C.T. (2007). Introduction to Psychology. New York: McGraw-Hill Publishing Company Limited.

উদ্দিন, আফসার (২০০২)। সাধারণ মনোবিজ্ঞান, রাজশাহী: রাজশাহী বিশ্ববিদ্যালয় প্রকাশন বোর্ড।

ছিদ্দিক, আবু বক্কর (২০২৩)। সাধারণ ও পরীক্ষণ মনোবিজ্ঞান, ঢাকা: প্রত্যয় পাবলিকেন্স।

মন্ডল, যোগেন্দ্র কুমার (২০০৮)। সাধারণ মনোবিজ্ঞান, ঢাকা: হাসান বুক হাউস।

সাইফুদ্দীন, কাজী (২০০৭) । সাধারণ মনোবিজ্ঞান, ঢাকা: আবির পাবলিকেসন্স।

সরকার, নিহাররঞ্জন (১৯৯০)। মনোবিজ্ঞান ও জীবন, ঢাকা: জ্ঞানকোষ প্রকাশনী।

হক, মোঃ আমিনুল (২০১৯)। সাধারণ মনোবিজ্ঞান ও পরীক্ষণ, ঢাকা: হাসান বুক হাউস।

Course Code	213403	Marks: 100	Credits: 4	Class Hours: 60
Course Title	Experimental Psychology			

Course Description

The core branch of psychology which is concerned with the scientific investigation of the individuals' responses to stimuli in situations is called Experimental Psychology. This subfield of psychology explores, and seeks to better understand behavior through the lens of empirical research methods. This course is designed primarily to provide students with knowledge about the principles, methods, and designs of psychological experiments.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	Understand the principles of scientific methods, the role of science in psychology, and the various types of scientific inquiry in psychological research
CLO2	Identify and apply the various research methodologies, including designing experiments, identifying variables, and conducting hypothesis testing
CLO3	Analyze experimental designs, including factors like control, sampling, and psychophysical methods, and demonstrate effective communication through writing experimental reports
CLO4	Apply ethical principles and guidelines in experimental research, particularly when working with human participants or animal subjects in psychology

Mapping CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	✓	✓	✓					✓				
CLO2		✓			✓			✓	✓	✓	✓	
CLO3				✓		✓	✓	✓				✓
CLO4						✓	✓					✓

Course Contents

Chapter	Topics	Teaching-Learning Strategies	CLOs
1	**Psychology and Science:** Science and commonsense, Goals or objective of science, Rules of science, Scientific method and its characteristics, Classification of science, Psychology as a science	Lecture (PPT), Group discussion, Concept mapping, Q&A	CLO1
2	**Nature and Scope of Experimental Psychology:** Experimental psychology, its scope, characteristics of experimental method, experiment versus systematic observation	Lecture, 3D model demonstration, Group activity, Video analysis, Q&A	CLO1, CLO2
3	**Variables:** Nature and types of variables, Independent, dependent, and intervening variables, Types of relationships between different variables studied in psychology, Measures of dependent variables, Nature and meaning of control, Control of independent variables, Extraneous variables	Interactive lecture, Role-play (on inter-sphere interaction), Field observation, Group discussion	CLO2, CLO3
4	**Research Problem:** Nature of a research problem, Sources of a research problem, Selection of a research problem	Interactive lecture, Group Discussion, PPT	CLO2, CLO3
5	**Hypothesis:** Definition and types of hypothesis, Sources of hypothesis, Functions of hypothesis, Importance or functions of hypothesis, Criteria of hypothesis	Lecture, Group Discussion, Presentation, Question-answer sessions.	CLO2, CLO3

6	**Designing and Conducting Experiments:** Types of experiments, Relationship studies in psychological experiments, Steps in planning an experiment, Conducting an experiment	Lecture, Group project on ecosystem system models, Simulation videos, Field trip/case visit.	CLO2, CLO3
7	**Experimental Designs:** Importance of design in research, Basis of selecting a design, Two independent groups design, Multiple groups design, Two matched groups design, Repeated measurement design, Factorial design, co-relational design	Lecture, Diagram drawing workshop, Model demonstration, Group poster presentation	CLO2, CLO3
8	**Psychophysical Methods:** Basic concepts of psychophysics, Historical background of psychophysics, Founders of Psychophysics-Weber and Fechner laws, Method of limit, method of constant stimuli, Method of average error	Lecture, Animated video, Concept mapping, Field-based discussion (aquifer visit or case study)	CLO2, CLO3
9	**Writing Experimental Reports:** Writing style, Organization of the report, The use of headings, Citing and referencing sources	Lecture, Group discussion, Peer review, Report writing workshops	CLO1, CLO3
10	**Ethics of Experimental Research:** Research with human participants, Ethical principles for animal research	Lecture, Case study analysis, Group discussion, Debate on ethical dilemmas	CLO4

List of Books

Anderson, B. F. (1969). *The Psychology of Experiment: A Introduction to Science Method.* Belment California: Books/Cole publishing company.

Andreas B. G (1967). *Experimental Psychology.* New York: John Wiley and Sons Inc. Brown..

C.W. and Ghiselli. E.F. (1955). *Scientific Methods in Psychology*. New York: McGraw Hill.

McGuigan. F. J. (2011). *Experimental Psychology: Methods of Research*. New Jersey: Prentice Hall.

উজ্জামান, মুহাম্মাদ আকরাম (২০২৩) । *পরীক্ষন মনোবিজ্ঞান,* ঢাকা: প্রত্যয় পাবলিকেসন্স।

ছিদ্দিক, আবু বক্কর (২০২৩) । *সাধারণ ও পরীক্ষণ মনোবিজ্ঞান,* ঢাকা: প্রত্যয় পাবলিকেসন্স।

সাইফুদ্দীন, কাজী (২০০৮) ।*পরীক্ষন মনোবিজ্ঞান ও গবেষণা পদ্ধতি,* ঢাকা: আবীর পাবলিকেসন্স।

হক, মোঃ আমিনুল (২০১৯) ।*সাধারণ মনোবিজ্ঞান ও পরীক্ষণ মনোবিজ্ঞান,* ঢাকা: হাসান বুক হাউস।

Course Code:	213405	Marks: 100	Credits: 4	Class Hours:60
Course Title:	Statistics in Psychology I			

Course Description

This course introduces students to the fundamental principles and techniques of statistics with a focus on their application in psychology. It covers the basic concepts of data collection, classification, graphical representation, and descriptive statistical analysis. Students will learn how to compute and interpret measures of central tendency, variability, skewness, kurtosis, correlation, and regression. The course emphasizes the importance of statistical thinking and its role in psychological research, enabling students to make informed decisions based on data and to interpret psychological phenomena through a quantitative lens.

Course Learning Outcomes (CLOs):

After going through the course, students will be able to learn the following outcomes:

CLO1	Demonstrate a comprehensive understanding of the fundamentals of statistics, including the nature, scope, and types of data, variables, and scales of measurement in psychological research
CLO2	Analyze and interpret frequency distributions, measures of central tendency, and variability in psychological data, and apply appropriate statistical tools for data analysis
CLO3	Apply advanced statistical concepts such as correlation, regression, and transformations in the context of psychological data, demonstrating the ability to select and use the correct statistical methods for analysis
CLO4	Critically assess and interpret statistical results in psychology, including the evaluation of skewness, kurtosis, and the appropriateness of statistical measures in research

Mapping CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	✓		✓	✓			✓				
CLO2			✓	✓		✓		✓		✓	✓	
CLO3		✓			✓	✓	✓	✓	✓			
CLO4					✓			✓	✓	✓		✓

Course Contents

Chapter	Topics	Teaching-Learning Strategies	CLOs
1	**Introduction:** Nature, scope definition and classification of statistics, Why statistics in psychology, Data and their types, The idea of quantification, scales of measurement, Variable and their types, Nature of psychological research	Lecture (PPT), Group Discussion, Q&A, Concept Mapping, Short Quiz	CLO1
2	**Symbolic Representation of Data:** Advantages of symbol in mathematics, Conventional rules for symbolic representation of data	Lecture, Thematic Maps Analysis, Case Study, Group Discussion, Multimedia Demonstration	CLO1, CLO2
3	**Frequency Distribution:** Grouping of data, Choosing class intervals, Apparent and true limits of class intervals, Frequency and cumulative frequency distributions, Graphic representation of data: histogram, frequency polygon and cumulative frequency polygon	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO2, CLO3
4	**Measures of Central Tendency:** What is central tendency? Computing mean, median, and mode, Properties and uses of the three measure of central tendency, When to apply mean, median and mode	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO2, CLO3
5	**Measures of Variability:** Need to measure variability, Computing range, inter-quartile range, semi-interquartile range, mean deviation, the variance, standard deviation and coefficient of variation	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO2, CLO3

6	**Measures of Skewness and Kurtosis:** Moments about the mean, Positive and negative skewness, Leptokurtic mesokurtic, and platykurtic distribution	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO2, CLO3
7	**Correlation:** Positive versus negative correlation, linear versus non-linear correlation, low, moderate, strong, and perfect correlation, The scatter diagram, Bivariate correlation: Pearson product-moment correlation, Spearman rank order correlation, Kendall's rank-order correlation, Pearson biserial and point biserial correlations; partial and semi-partial (or part) correlations	Lecture, Case Studies, Data Analysis Activities, Group Work	CLO3, CLO4
8	**Regression:** Some important information about straight lines, The method of least squares, Assessing the goodness of fit: sum of squares, *R* and R^2, Assessing individual predictors, Types of regressions	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO1, CLO2
9	**Score Transformations:** The need for transformed/derived scores, Types of transformations: linear transformation- *z* and *T* scores, non-linear transformation – percentile points and percentile ranks, normalized transformation- stanines, Combining measures from different distributions	Lecture, Group Activities, Practical Demonstrations, Discussion	CLO3, CLO4

List of Books

Blommers, P. & Lindquist, E. F. (1960). *Elementary Statistical Methods in Psychology and Education.* New Delhi: Book Company.

Ferguson, G. A. & Takane.Y. (1989). *Statistical Analysis in Psychology and Education.* NY: McGraw Hill Book Company

Garrett, H. F. (2012). *Statistics in Psychology and Education.* Kamla Nagar: Surjeet Publications.

Guilford, J. P. (1979). *Fundamental Statistics in Psychology and Education*. NY: McGraw Hill.

Minium, E. W., King, B. M., & Bear. G. (2004). Statistical Reasoning in Psychology and Education. Singapore: John Wiley & Sons, Inc.

আহমদ, মোঃ নাজির ও রহমান, মোঃ মিজানুর (২০২১)। *মনোবিজ্ঞানে পরিসংখ্যা-১*, ঢাকা: প্রত্যয় পাবলিকেশন্স।

তাহের, মোঃ আবু ও হক, মোঃ আমিনুল (২০২১)। *মনোবিজ্ঞানে পরিসংখ্যান*, ঢাকা: হাসান বুক হাউস।

সরকার, বৈদ্যনাথ (২০২৩)। *মনোবিজ্ঞানে পরিসংখ্যান প্রথম খণ্ড*, ঢাকা: প্রত্যয় পাবলিকেশন্স।

Course Code:	**213406**	**Marks: 100**	**Credits: 4**	**Class Hours: 60**
Course Title:	**Practical: Introduction to Psychology**			
Marks Distribution	Laboratory Note Book 10x1= 10			
	Conduction 20x1= 20			
	Report/Script 50x1= 50			
	Viva 20x1= 20			

Course Description

This course provides hands-on experience in designing, conducting, and analyzing psychological experiments. Students will explore foundational psychological processes through laboratory-based activities focusing on learning, memory, reaction time, attention, motor performance, and sensory perception. The course aims to develop students' practical skills in applying experimental methods and using scientific techniques to study human behavior.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	Understand and apply the principles of estimating time under both filled and unfilled conditions in experimental settings
CLO2	Identify and classify different types of variables used in psychological research
CLO3	Investigate the role of stimulus orientation in visual short-term memory and its impact on memory retention
CLO4	Design and conduct basic psychological experiments using appropriate tools and methods

Mapping CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	✓			✓			✓				
CLO2	✓	✓	✓		✓		✓	✓	✓	✓	✓	✓
CLO3	✓	✓	✓		✓			✓	✓			✓
CLO4	✓	✓	✓		✓	✓		✓			✓	✓

Course Contents

Module No.	Contents	Teaching-Learning Strategies	CLOs
1	Estimation of time under filled and unfilled conditions	Lecture with Demonstration, Guided Drawing Practice, Q/A Session, Class Discussion	CLO1, CLO2, CLO3
2	Effect of feedback on sensory motor performance	Group Work, Hands-on Exercise, Field Demonstration, Use of Graph Paper	CLO2, CLO4
3	Effect of stimulus orientation on visual short-term memory	Lecture, Group Discussion, Experimental, Assignment	CLO1, CLO3
4	To study the retention as a function of time	On-site Task, Visual Demonstration, Manual Exercise, Observation Task	CLO3, CLO4
5	To study the effect of division of attention on learning /performance	Experimental, Hands-on Activities, Group Discussions	CLO2, CLO4
6	To investigate the phenomenon of bilateral transfer of learning	Lecture, Experimental, Class Discussion	CLO4
7	To compare incidental versus intentional learning.	Lecture, Group Discussion, Experimental, Assignment	CLO2, CLO4

8	To study the effect of natural and emotional word on reaction time.	Lecture with Demonstration, Guided Drawing Practice, Q/A Session, Class Discussion	CLO3
9	To study the role of organization on verbal learning	Lecture, Experimental, Class Discussion	CLO2, CLO4
10	To study the effect of retroactive inhibition on paired-associate learning	Lecture with Demonstration, Guided Drawing Practice, Q/A Session, Class Discussion	CLO1, CLO3

List of Books

Baddeley, A. D. (1986). *Working memory*. Oxford: Oxford University Press.

Baddeley, A. D. (2000). *The episodic buffer: A new component of working memory?* Trends in Cognitive Science 4, 417-423.

Ebbinghaus, H. (1985). *Memory: A Contribution to Experimental Psychology*. New York: Colombia University

Hull, G. L. (1933). *The Meaningfulness of 320 selected Nonsense Syllables.* Amer J. Psy 45, 730- 734.

রহমান, মো. আবদুর (২০১৪)। *উচ্চতর ব্যবহারিক মনোবিজ্ঞান,* বগুড়া: প্রকাশিকা সা.রহমান।

সাহা, নিতাই কুমার (২০১৯)। *হাতে কলমে মনোবিজ্ঞান প্রথম খন্ড*, ঢাকা: হাসান বুক হাউস।

Course Code:	219901	Marks: 100	Credits: 4
Course Title:	History of Bangladesh: Language, Culture, and Identity		Hours: 60

Course Objective:
This course is designed to help undergraduate students from diverse academic backgrounds develop a comprehensive and nuanced understanding of the historical development of Bangladesh, with a particular focus on its language, culture, and identity. By critically examining a variety of historical events, socio-political movements, and cultural shifts from ancient times to the contemporary period, students will have the opportunity to trace the evolution of the Bengali nation. The course aims to foster informed citizenship through an exploration of the Liberation War, identity formation processes, cultural heritage, the lives and contributions of key political figures, and the roles of ethnic minorities. Ultimately, the objective is to equip learners with the intellectual tools necessary to contextualize current national debates and to recognize the role of youth and globalization in shaping the future of Bangladesh.

Course Learning Outcomes:

At the end of the course, learners will be able to:
CLO 1: Recall key historical events, dates, movements, personalities, and cultural developments that contributed to the emergence of Bangladesh from the pre-colonial period to the present day.

CLO 2: Demonstrate an understanding of the socio-political, linguistic, and cultural influences that shaped Bengali identity and nationalism, emphasizing the Language Movement, Liberation War, and subsequent democratic uprisings.

CLO 3: Apply their historical and cultural understanding to interpret contemporary national issues and debates relating to identity, political reform, and cultural transformation in Bangladesh.

CLO 4: Analyze how historical events, political ideologies, and social changes are interconnected in shaping the current socio-political landscape of Bangladesh, with a particular emphasis on the roles of movements, minorities, and the youth.

Course Contents

Course contents	Teaching Learning Strategy	CLOs
1. Pre-colonial Era • Life and Culture of the People in Ancient Bengal • Bengal Under Muslim Rulers: Society, Culture, and Religion • Role of the Sufis in Preaching Islam and Impacts of • Sufism on the Bengali Society • Bengali Society and Culture in the Writings of Foreigners	Interactive lectures	CLO 1,2
2. Colonial Era (18th and 19th Century) • The Battle of Plassey (1757) and the Beginning of British Colonialism • Bengal Renaissance • Reforms in Hindu Society • New Forms in Bengali Literature and Culture • The Muslim Response to Western Education	Interactive lectures, reading, and assignments	CLO 1,2
3. Colonial Era (First Half of the 20th Century) • Partition of Bengal (1905) • Hindu-Muslim Disagreements • Formation of the All India Muslim League (AIML) • Muslim Shahitya Samaj • Buddhir Mukti Andolan: The Urge for Rational Thinking in Bengali Muslim Society • Growth of Religion-based Identity • Politics of Hindutva and the Two-Nation Theory • Spread of Communalism in Society • The Partition of India and Bengal	Interactive lectures and group discussions	CLO 2,3
4. Post-Partition Era (1947-1971) • Language Movement • Political, Economic, and Cultural Aspects • Growth of Vernacular Nationalism • Cultural Activism	Interactive lectures and group discussion	CLO 2,3
5. Changing Bengali Identity • The Evolution of Bengali Identity in the Context of Language, Culture, and Religion • From Ancient Times to the Present • The Role of the Bengali Language in Shaping Identity • The Language Movement of 1952 and Its Long-term Impact on National • Consciousness • The Influence of Religion on Bengali Identity • The Impact of Socio-political Movements on Identity	Interactive lectures and thematic assignments	CLO 2,3

Formation • The Liberation War of 1971 • Contemporary Debates on Bengali Identity • The Role of Youth in Redefining Identity • The Influence of Globalization		
6. Liberation Movement of 1971 and Mass Uprising till 2024 • The Political and Economic Exploitation of East Pakistan by West Pakistan • Six (6) Points Movement, Uprising of 1969 • The 1970 General Elections • The Non-cooperation Movement and the Declaration of Independence • The Role of AK Fazlul Haque, Huseyn Shaheed Suhrawardy, Maulana Bhashani, Sheikh Mujibur Rahman and Ziaur Rahman • The Liberation War of 1971 • Genocide and Resistance • The Role of the Mukti Bahini • Post-independence Challenges • Nation-building • Political Instability • Mass Uprisings and Democratic Movements • The Anti-autocracy Movement of the 1980s • The 1990s Movement for Democracy and Afterwards • The 2024 Movements for Political and Social Reforms	Interactive lectures, group discussions, and thematic assignments	CLO 1 CLO 3-4
7. History of Other Ethnic Groups • The Indigenous Communities of Bangladesh • Historical Presence • Cultural Practices • Land Rights and Cultural Assimilation • Contributions and Sacrifices • Political Representation of Ethnic Minorities	Interactive lectures and group discussions	CLO 4
8. Cultural Heritage and Modern Transformations • The Evolution of Bengali Culture • From Ancient Traditions to Modern Expressions • The Role of Literature, Music, and Art in Shaping Bengali Identity • Contributions of Rabindranath Tagore, Kazi Nazrul Islam, and Other Cultural Icons • The Impact of Globalization on Bengali Culture • The Revival of Traditional Arts • The Role of Youth in Cultural Innovation Urbanization • Environmental Changes	Interactive lectures, documentary screening, and thematic assignments	CLO 1-4

• The Commodification of Culture		

List of Books

Ahmed, M. (1979). *Bangladesh: The constitutional quest for autonomy.* Dhaka: University Press Limited.

Ahmed, R. (1981). *The Bengal muslims 1871-1906: A quest for identity.* Oxford University Press.

Alavi, H. (1972). *The state in post-colonial societies: Pakistan and Bangladesh.* In K. Gough & 11. P. Sharma (Eds.), Imperialism and revolution in South Asia (pp. 145-178). New York, NY: Monthly Review Press.

Bernier, F. (2023). *Travels in the Mogul Empire: A.D. 1656-1668.* Chennai: Atlantic Publishers.

Bleie, T. (2005). *Tribal peoples, nationalism, and the human rights challenge.* The Adivasis of Bangladesh. University Press Limited.

Bosc, S. (2011). *Dead reckoning memories of the 1971 Bangladesh war*. Hurst & Company.

Eaton, R. M. (1996). *The rise of Islam and the Bengal frontier. 1204-1760.* Berkeley: University of California Press.

Gilmour, D. (2019), *The British in India: Three centuries of ambition and experience*. London. Penguin.

Habib, L. (1982). *Cambridge economic history of India. Cambridge.* Cambridge University Press.

Halim, S., Amanullah, A. S. M., & Nasir, R. 1. (Eds.). (2024). *Society and sociology in Bangladesh*: *A South Asian perspective.* The University Press Limited.

Hashmi, T. (2021). *Fifty Years of Bangladesh, 1971-2021 Crises of Culture, Development, Governance and Identity.* Switzerland: Palgrave Macmillan.

Huq, M. E. (1975). *History of Sufism in Bengal. Dhaka.* Bangladesh Asiatic Society

Husain, 1. (2014). *Karl Marx on India.* New Delhi: Tulika Books.

Jahan, R. (1972). *Pakistan: Failure in national integration.* New York, NY: Columbia University Press.

Maniruzzaman, T. (1988*). The Bangladesh revolution and its aftermath.* (2nd ed.). Dhaka: University Press Limited. (Original work published 19801

Mascarenhas, A. (1986). *Bangladesh: A legacy of blood. London:* Hodder and Stoughton.

Rashid, H. O. (2015). *The foreshadowing of Bangladesh: Bengal Muslim League and Muslim politics* (3rd ed.). Dhaka: University Press Limited.

Roy, A. (1984). *The Islamic syncretistic tradition in Bengal. Princeton.* NJ: Princeton University Press.

Sen, A. (2006). *Identity and violence: The illusion of destiny.* W. W. Norton & Company.

Umar, B. (2022). *The Emergency of Bangladesh: A History of East Pakistan.* Dhaka: Bangla Gobeshona.

Course Code	219903	Marks: 75	Credits: 3
Course Title:	Information and Communication Technology		

Course Objectives:

The main objective of the course is to develop students' understanding and skills in using, managing, and applying technology to solve problems and enhance various aspects of life and work. Key areas of focus include understanding ICT systems, software, hardware, networks, and their applications.

Course Learning Outcomes: The student will be able to:

CLO 1	Explain foundational ICT concepts, including the information processing cycle.
CLO 2	Operate standard computer hardware and software systems effectively.
CLO 3	Use office productivity tools (Word, Excel, PowerPoint) for academic and professional tasks.
CLO 4	Apply safe internet practices and use internet tools for communication and information retrieval.
CLO 5	Understand and explain the concepts of Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML)
CLO 6	Understand the concept of free-lancing

Course Contents:

Unit	Specific Objectives	Content	Teaching and Learning Approach
Unit 1: Introduction to Information and Communications Technology (ICT) and Computer System	The student will be able to: • Explain the concept of ICT and its related terminologies. • Describe the information processing cycle. • Analyze the impact of ICT on educational, social and economic development. • Identify career opportunities in ICT education. • Differentiate among the	Definition of ICT, basic concepts and terminologies. Data, Information Application of ICT, Advantages and Disadvantages of ICT, Impact of ICT Career opportunities in ICT education Information Processing Cycle. Classification of Computers The Vital Components of the Systems Unit.	Classroom Lectures

	classes of computers and the usages of them. • Identify the vital components of the Systems Unit.		
Unit 2: Computer Hardware and Software	The student will be able to: • Describe the categories of Computer Hardware. • Describe the commonly used Input and Output devices • Identify the main processing devices, storage devices and media. Identify the main communication devices. • Identify types of Software packages. • Distinguish between an Operating System and Application software. • Identify different types, examples and uses of Operating Systems and Application software • Distinguish between Open Source and Proprietary Software.	Categories of Computer Hardware: Input devices, Processing devices, Output devices, Storage devices Communication devices, Main Processing Devices: The Processor, Control Unit and Arithmetic and Logic Unit Software Packages, Operating Systems, Types and uses of Operating Systems, Types and uses Application Software: Educational software, Games software, Graphics software Browsers: Internet explorer, Google chrome, Mozilla Firefox, Opera, Internet explorer, Mozilla Firefox; Proprietary and Open Source Software	Classroom lectures and Lab
Unit 3: Introduction to Word Processing Application	• The student will be able to: • Identify Word Processing packages. • Create and save a document using the Word Processor. • Format a Word document using formatting tools. Demonstrate the ability to perform collaborative editing. • Insert tables in a Word Processing document. • Insert symbols and pictures in Word	Word Processing Packages and Their Uses. Introduction of Microsoft Word Software. File creation and saving: New, Save, Open, Save As. Editing: Copy/Cut, Paste, Undo, Redo, Find, Replace, Clipboard, Sub-Headings And Paragraphs, Text Correction, Wrapping Text, Text Orientation **Formatting:** Font: Style, Size, Color, Bold, Underline, Italic, Superscript,	Classroom Lectures, Lab, and Hands-on Practice

	Processing documents. • Use layout techniques in document creation. • Inserting headers and footers. • Print documents using the various print options.	Subscript, Shadow, Strikeout. Paragraph: Alignment, Bullet & Numbering, Indent, Line Spacing, Table Border **Insert:** Table, Picture, Shape, Smart Art, Chart, Header, Footer, Page Number, WordArt, Drop Cap, Symbols, Equations, Hyperlink, Comment. **Page Design:** watermark, page border **Layout:** Page setup, print	
Unit 4: Spreadsheet Application	• The student will be able to: • Identify Spreadsheet Packages. • Explain the importance of the Spreadsheet application in data management. • Explain related concepts and terminologies in the Spreadsheet. • Identify features in the Spreadsheet application window. • Create and save a Workbook. • Construct and insert simple formulae and functions. • Format the worksheet using formatting tools. • Printing a worksheet.	Spreadsheet Packages, e.g. Excel. Importance of Spreadsheet application in Data Management, Related Concepts and Terminologies (e.g., cell(s), rows, columns, worksheet, workbook) Excel Tool Bars: Formatting Bar, Standard Bar, Formula Bar. Types of Data and Their Uses (e.g., number, date, text, currency), Creating and Saving a Workbook, Formula and Functions, Sorting, filtering, validation, creating pivot table. Formatting Worksheet Using Formatting Tools. Draw Graph/chart, Printing Worksheet.	Classroom Lectures, Lab and Hands-on Practice
Unit 5: Presentation Application	• The student will be able to: • State the importance of the Presentation application. • Identify the commonly used features of a Presentation application while prepare a presentation.	Presentation Applications Packages, Devices used for Presenting, Importance of Presentation Application, Principles for Designing Presentations, Terminologies in Presentation Application (eg, , Slide transitions, Slide show, etc.),	Classroom Lectures, Lab and Hands-on Practice

	• Create and save presentations using a template, Add new slide(s). Edit text, Format text, Insert objects, images and pictures, Run slide show, Apply transition, animation effects to slides • Select the print option for printing. • Prepare a presentation on a selected topic and present it.	Commonly Used Features of Presentation Application Window: Toolbars, Different presentation view modes Prepare a Presentation: Slide creating, editing, theme, background, inserting elements (image, audio, video hyperlink, table, smart art, graph, etc.) and formatting slides, Slide Layout, slide Show, slide transition, animation Effects. Selection of Print Option: Entire presentation, Specific slides, Handouts, Notes pages, outline view of slides.	
Unit 6: Privacy and Security	The student will be able to: • Understand the basics of digital security • Use some security tools. • Understanding digital ethics.	Introduction to Information Security, cybercrime, DoS and DDoS Attack, Key Management, Digital Signature and Certifications, privacy, Data Security, Vulnerability, Threat and Risk, Malware, Social Engineering, Hacking, Plagiarism, Fishing, Software Piracy, Worms and Viruses, Spam, Adware, Malware, Spyware, Antivirus Software Ethics in the digital world	Classroom Lectures, Lab and Hands-on Practice
Unit 7: Using The Internet to Communicate and Accessing Information	• The student will be able to: • Explain basic concepts, requirements, and terminologies of the Internet • Apply the rules and regulations in the use of the internet. • Using email • Use the internet social network to communicate. • Use Uniform Resource Locators (URLs) to	Internet, Intranet, Extranet, IP Address, Masking, MAC Address, Internet Services, OSI Reference Model, TCP/IP protocol stack, IPv4, IPv6, subnet Masking, MAC Address, Internet Services, Network Configuration and Troubleshooting, Wi-Fi, Broadband, Email Usage. Rules and Regulations in the Use of the Internet: Spam-Unsolicited Emails, People's Privacy, Intellectual Property Rights, etc.	Classroom Lectures, Lab and Hands-on Practice

	access Information. Use search engines to access information • Upload files to virtual drives and work on it.	E-mail: Creating an Email Account, Sending, Accessing Email Messages, Attaching Documents to Email Messages, Using the Internet: Social Networks to Communicate, Uniform Resource Locators (URLs) to Access Information, Using Search Engines, Downloading Information from the Internet. Transferring Information from the Internet to a Different Application Intellectual Property Rights, ICT Policy, Software Piracy etc. Using Cloud Space: Google Drive, Google Workspace, OneDrive, Dropbox, etc.	
Unit 8: Emerging Technologies: Data analytics, Artificial intelligence, Machine learning	• Define Data Analytics, Artificial Intelligence (AI), and Machine Learning (ML). • Understand how data is collected, processed, and used for decision-making. • Recognize the role of AI and ML in everyday life. • Explain basic differences between AI and ML. • Discuss benefits and challenges of these technologies. • Recognize current trends and career opportunities in these fields.	**Introduction to Data Analytics:** Define Data, Types of Data, Define Data Analytics, Basic Steps in Data Analytics, Applications of Data Analytics, Common Tools of Data Analytics (Excel, Tableau, Power BI, Python, SQL, etc.). **Artificial Intelligence (AI):** Define AI, Objectives of AI, AI capabilities, AI applications (NLP, Image Recognition, Predictive Analytics, Recommender Systems, Robotics, etc.); **AI Tools**: OpenAI GPT, TensorFlow, PyTorch, Keras, Scikit-learn,etc.), AI in daily life (Voice assistants, facial recognition, GPS route suggestions, chatbots, etc.) .	Classroom Lectures, Lab and Hands-on Practice

		Machine Learning (ML): Define Machine Learning (ML), How It Works, Classification of ML, applications of ML, and common ML platforms (Teachable Machine, TensorFlow, PyTorch, Microsoft Azure Machine Learning, etc.). Relationship Between Data Analytics, AI, and ML Benefits & Challenges of Data Analytics, AI, and ML. Future Trends & Career Paths.	
Unit 9: Introduction to Freelancing	The student will be able to: • Define and explain the fundamental concept and significance of freelancing in the global context. • Identify and categorize various online marketplaces and professional fields. • Develop professional profiles, portfolios, and proposals suitable for freelancing platforms. • Demonstrate the use of online payment systems and financial transaction methods. • Analyze and evaluate strategies for career development and effectively apply techniques to overcome freelancing challenges.	Introduction and Basic Concepts, Fields of Freelancing, Essential Skills and Preparation, Introduction to Freelancing Marketplaces, Creating Profile and Portfolio, Earning and Transaction Process, Building a Freelancing Career, Challenges and Solutions.	Classroom Lectures, Practical Demos, Assignments, Discussions

List of Books:

Norton, P. (2008). *Introduction to computers* (9th ed.). The McGraw-Hill.

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft Office 2016 step by step*. Microsoft Press.

Hassan, N. A., & Hijazi, R. (2017). *Digital privacy and security using Windows: A practical guide*. Apress.

Maheshwari, A. (2024). *Data analytics made accessible*. Jay Cobb.

Russell, S. J., & Norvig, P. (2020). *Artificial intelligence: A modern approach* (4th ed.). Pearson.

Alpaydin, E. (2020). *Introduction to machine learning* (4th ed.). MIT Press.

Course Code	**219904**	**Marks: 25**	**Credit: 1**
Course Title:	**Lab: Information and Communication Technology**		

Course Objectives:

The main objective of the course is to teach the students' understanding and skills in using, and applying technology to solve problems and enhance various aspects of life and work. It includes assembling hardware, installing software, preparing PPT slides, and producing Word and Excel documents.

Course Learning Outcomes:

The student will be able to:

CLO 1	Learn assembling hardware
CLO 2	Prepare, edit and print Word documents and excel
CLO 3	Prepare Power Point presentation.
CLO 4	Access information from Internet, e-mail
CLO 5	Installation of anti-virus software
CLO 6	Data collection, Prediction, Pattern recognition, classification
CLO 7	Exploring markets, Proposal writing, Simulation strategies

List of Experiments

CLO Addressed	Unit	Experiments	Teaching Learning Approach
CLO1	1-4	• Assemble different hardware • Install different software • Operate the computer - Drive, folder and file management • Maintenance	Lab and Hands-on Practice
CLO 2	3	**Word** • Prepare a Word document on a specific topic (e.g, routine, question paper, CV, reports, applications) • Formatting the document (Alignment, table, border, watermark, etc.), e.g., newspaper article, academic report, or documentation used in daily life, book, poster • Print documents with different paper and printers	Lab and Hands-on Practice
CLO 2	4	• **Excel** • Prepare a grade sheet • Prepare a family expenditure • Prepare a business expenditure report • Prepare payroll management, with a report • Create graphs on the given data • Print Excel files	Lab and Hands-on Practice
CLO 3	5	**Power point** • Prepare an academic presentation on a specific topic. • Formatting the slides & using different tools. • Apply animation and transition • Print PPT files in different modes: Hand note, Slides shorter, outline	Lab and Hands-on Practice
CLO4	6	• Install antivirus software, e.g., Norton Antivirus, McAfee, Kaspersky, Avast.	Lab and Hands-on Practice
CLO5	7	1. Use of email 2. Access information from the internet, use a search engine. 3. Use of virtual drive for collaboration 4. Google Meet	Lab and Hands-on Practice
CLO6	8	• Data Collection and Visualization (Excel) • Simple Prediction (Excel) • Pattern recognition (Teachable Machine) • Classification (Teachable Machine)	Lab and Hands-on Practice

CLO7	9	**Free lancing** • Marketplace exploration and portfolio creation • Proposal writing • Payment simulation	Lab, Hands-on, Group Work, Discussion

Instructions for the Lab:

1. Lab sessions should be hands-on, with students engaging in tasks like assembling hardware, installing software, and using office tools.
2. Software like Microsoft Word, Excel, PowerPoint, and Teachable Machine will be used for document creation, data analysis, and basic machine learning.
3. Labs also include antivirus installation, email use, and online collaboration via Google Meet and Drive, emphasizing teamwork and problem-solving.

Instructions for the Lab exam:

1. The lab exam is individual. Students must answer two out of seven questions.
2. Marks distribution: Experiment (20 marks, 10 for each) for practical tasks, Viva-Voce (5 marks) for oral assessment. Total: 25 marks.

List of Books:

Comer, D. E. (2018). *The internet book: Everything you need to know about computer networking and how the internet works* (5th ed.). Chapman and Hall/CRC Press.

Lambert, J., & Frye, C. (2015). *Microsoft office 2016 step by step*. Microsoft Press

Course Code	**212009**	**Marks:100**	**Credits:4**	**Class Hours:60**
Course Title	**Introducing Sociology**			

Course Objectives

The course is designed to provide foundational knowledge of sociology as a social science. It will explain key concept, principles and perspectives in sociology. It will highlight various structural issues like major social institutions, social classes, inequality, population and urbanization. Besides this, the course helps the students to develop sociological thinking to understand society and human behavior.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes

CLO1	Students will be demonstrating knowledge about preliminary ideas of sociology
CLO2	They will be informed about the contributions of sociologists

CLO3	They will apply several procedures of scientific social research
CLO4	They will explain various terms of culture and cultural diversity in society
CLO5	Students will get knowledge about different types and evolution of society
CLO6	Students will know both about social institutions and various factors which contribute to social stratification and mobility within society
CLO7	They will recognize different social problems of Bangladesh
CLO8	They will be able to analyze the influences of social forces on individual behavior and social change which happens globally

Mapping of CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PL05	PLO6	PLO7	PL0-8	PLO9	PLO10
CLO1	✓									
CLO2	✓									
CLO3					✓					
CLO4	✓									✓
CLO5	✓	✓								
CLO6	✓									
CLO7				✓					✓	
CLO8	✓		✓				✓			

Course Contents

Chapter	Topics	Teaching-Learning Strategies	CLOs
1	**Definition, Scope and Development of Sociology:** Definition of Sociology, Nature and scope of Sociology, Emergence of Sociology, Importance of Sociology as a science, Value freeness of sociology, relation with other scientific subjects	Lecture and Assignment	CLO-1
2	**Founding thinkers and major theoretical perspectives:** Auguste Comte, Karl Marx, Herbert Spencer, Emile Durkheim, Max Weber, Functionalist, conflict and	Lecture and Assignment	CLO-2
3	**Major Methods and Techniques of Sociology:** Social research, Qualitative and Quantitative techniques, Survey	Lecture, Fieldwork	CLO-3

	method, Comparative Method, historical method, Scientific method, Case study, FGD, Interviews, Observation, and Experiments. Ethics in social research	and Group presentation	
4.	**Culture, Society and Civilization:** Meaning and characteristics of culture, elements of culture and cultural change, Cultural lag, civilization, culture and civilization, community, organization and group	Lecture, Assignment and Group Discussion	CLO-4
5	**Evolution and different types of society:** Hunting and food gathering society, Horticultural society, Agricultural Society, Industrial society, Primitive communism, Slavery, Feudal, Capital, Socialism, Different types of capitalism, Socialism and modern communism	Lecture and Assignment	CLO-5
6.	**Social Institutions:** Marriage and family, education, religion, economy and state. Characteristics and functions of each institution. Changing nature of institutions in modern society	Lecture and Assignment	CLO-6
7.	**Social problems and Inequality:** Nature of social problems, Population problem, poverty, unemployment, crime and deviance, corruption, drug addiction, over-urbanization. Social inequality and forms of stratification: caste, class race and gender. Theories of inequality and Social mobility	Lecture and Assignment	CLO-7
8.	**Social Interaction and Social Change:** Social interaction and exchange, socialization, Causes of Social Change, theories of Social Change, technology and Social Change, globalization and its impact. Postmodern and postindustrial society.	Lecture and Assignment and Group Discussion	CLO-8

List of Books

Giddens, A. (2009). *Sociology (6th ed.).* Cambridge, UK: The Polity Press.

Goldthorpe, J. E., & Goldthorpe, J. E. (1985). *An introduction to sociology*. Cambridge, UK.Cambridge, University Press.

Halim, S., Amanullah, A. S. M., & Nasir, R. 1. (Eds.). (2024). *Society and sociology in Bangladesh*: *A South Asian perspective.* The University Press Limited.

Haralambos, M., & Holborn, M.(2013). *Sociology:Themes and Perspectives*. London, UK: Harper Collins publications.

Ritzer, G. & Murphy, W. W.(2019). *Introduction to sociology*. London, UK: SAGE publications.

Schaefer, R., T. (2012). *Sociology: A Brief Introduction (10th ed.)*. New York, USA: McGraw-Hill.

Stolley, K. S. (2005). *The Basics of Sociology*. (1st ed).Boston, USA: Greenwood Press.

Turner, B. S. (1999*). Classical Sociology*.(1st ed). London,UK: Sage Publications.

সেন, রঙ্গলাল (১৯৯৭). *বাংলাদেশের সামাজিক স্তরবিন্যাস,*(২য় সংস্করণ), ঢাকা, বাংলাদেশ: নিউ এজ পাবলিকেশন।

Course Code: 212109	Marks: 100	Credits: 4	Credit Hours: 60	BSS Hon's 1st Year
Course Title: Introduction to Social Work (Allied Course)			Course Orientation: Theoretical	

Course Description

Introductory social work is an allied course for the registered students of respective disciplines. The focal objective of this course is to spread basic knowledge of social work education along with the concept, methods, relationship of social work with other disciplines, practice in different settings and social work as a profession. The course also covers industrial revolution, the philosophical foundations of social work, the development of professional education and its relationship with various religious creeds. Ethical principles and values are also discussed in this course.

Course Objectives

The objectives of this course encompass:

- To familiarize students about historical development and associated disciplines of social work;
- To help students able to understand the social work profession and industrial revolution;
- To attain knowledge about major social problems and intervention of social work;
- To introduce students about social work methods, types and practice in different settings;
- To introduce students to research and social research in social work arenas;
- Introduce students to religious and philosophical foundations of social work.

Course Contents

Chapter	Topics	Teaching Learning Strategy	CLOs
1	**Concepts Relevant to Social Work Practice:** Social Work, Social Welfare, Social Value, Social Reform, Social Legislation, Social Security, Social Development, Good Governance, Climate Justice, Field Work.	Class Lectures, PP Presentations, Content Analysis	CLO1
2	**Associated Disciplines and Their Relations with Social Work:** Sociology, Psychology, Economics, Political Science, Law, Journalism, Medical and Public Health, Anthropology.	Lesson Plans, Lectures, Handouts, Paraphrasing	CLO1

3	**a. International Growth and Development of Social Work:** UK and USA. **b. Social Welfare Practices and Movements in Undivided India and Bangladesh:** Traditional Welfare Activities and Charity, Social Reformers and their Movements. **c. Evolution of Social Work Education in the Sub-continent and Bangladesh.**	Lectures, Handouts, Exhibition of Audio-Visual Recordings, PP Presentations.	CLO2
4	**Social Work Profession**: Profession, Occupation, Social Work as a Profession, Values and Professional Code of Ethics of Social Work, NASW.	Lectures, Seminar Presentation, Group Discussion.	CLO2
5	**Social Problems and Social Services in Bangladesh:** Major Social Problems and Intervention of Social Work, Social Work Practice in Govt. Organizations (USS, RSS) and NGOs, Probation and Parole.	Lectures, Content Analysis, Visit And Demonstration	CLO3
6	**Industrial Revolution (IR):** Renaissance, Globalization, Industrialization, Urbanization, Impacts of IR on Society, Facing 4th IR challenges with Communication skill, Analytical Ability, Problem Solving Ability, Digital Literacy and Adaptability; Skills of Social Work,5th Industrial Revolution and Social Work.	Lectures, Handouts, Audio-Visual Recordings, Visit And Demonstration. symposium,	CLO4
7	**Types and Methods of Social Work:** Types, Green Social Work, Basic and Auxiliary Methods; Concept, Elements, Principles, Significance of Social Work Methods in Bangladesh.	Lectures, Content Analysis, Seminar presentation, Group Discussion	CLO5
8	**Social Research:** Research, Research as a Scientific Method, Social Research, Methods, Techniques and Tools of Social Research, Sampling and Its Types, Research Proposal, Importance of Social Research.	Lectures, Handouts, Reading, Assignments, Content Analysis	CLO5
9	**Social Philosophy and Religion:** Social Philosophy, Individualism, Socialism, Welfare State, Contribution of Major Religions on Social Work and Humanity, Philosophical, Religions and Humanitarian Basis of Social Work.	Lectures, PP presentations, Workshop etc. according to pedagogy.	CLO6

Course Learning Outcomes (CLOs)

After Fruitful conclusion of the course apprentices can be able to:

CLO1: Know about the related concepts and associated disciplines of social work;
CLO2: Elucidate the international growth and development of social work profession;
CLO3: Demonstrate major social problems and social services in Bangladesh;
CLO4: Understand the Industrial revolution and its challenges;
CLO5: Know about social work types, methods of social work and social research;
CLO6: Understanding of social philosophy, religions and social work.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	•		✓		•		✓	✓		•	✓
CLO2	•	✓	✓		✓	✓		•		✓	✓	
CLO3	✓		•				✓		•			✓
CLO4		✓	✓	✓		•		✓	✓	✓	✓	
CLO5	•				✓	✓		✓				✓
CLO6	✓	✓	•		✓			✓	✓	•	✓	•

✓= Strong Relation	• =Weak Relation	Blank = Limited or No Relation

List of Books

Barker, R. L. (1995). *The Social Work Dictionary*. Washington DC: NASW Press.

Friedlander, W.A. (1977). *An Introduction to Social Welfare*. New York: Prentice Hall.

Houssian, M.N. & Alauddin, M. (1990). *Introduction to Social Work Methods*. Dhaka: Bangladesh, Anwari publication.

Morales, A. & Sheafor, B. W. (1977). *Social Work: A Profession of Many Faces*. Boston: USA, Allyn and Bacon.

Skidmore, R. A. & others. (1993). *Introduction to social work.* New Jersey: USA, Prentice Hall.

ইসলাম, ড. মো. নুরন্নল (২০১৪)। *সমাজকর্ম পরিচিতি*, ঢাকা: বাংলাদেশ, তাসমিয়া পাবলিকেশন্স।

ইসলাম, মো. শহিদুল (২০০৮)। *সমাজকর্ম*, ঢাকা: বাংলাদেশ, গ্রন্থ কুটির।

উদ্দিন, আহমেদ আয়েশ (১৯৭৮)। *রাষ্ট্রচিন্তা পরিচিতি প্লেটো থেকে মার্ক* (২য় সংস্করণ), ঢাকা: বাংলাদেশ, আইডিয়াল লাইব্রেরী।

করিম, রেজাউল (২০০৬)। *সমাজকর্ম: ধারণা, ইতিহাস ও দর্শন*, ঢাকা: বাংলাদেশ, জ্ঞানকোষ প্রকাশনী।

রহমান, হাবিবুর, দাস, তুলসী, ইসলাম, মো. ফখরন্নল ও ইসলাম, মো. রেজাউল (অনূদিত) (১৯৯৭)। *সমাজকর্ম অনুশীলন*, ঢাকা: বাংলাদেশ, গতিধারা।

Course Code: 211909		Marks: 100	Credits: 4
Course Title:	Introduction to Political Science		

Course Description

Introduction to Political Science introduces students to the fundamental concepts of state, government, law, rights, and political institutions. The course explores different systems of governance, the role of constitutions, political parties, and public opinion, while linking these with human behavior and social psychology. It helps psychology students understand how political structures influence attitudes, decision-making, and collective action in society.

Course Objectives (COs)

CO 1	To absorb the basic concepts of political organization and their composition, functions and institutional structures;
CO 2	To identify various forms of government and organs of the state;
CO 3	To identify and explain the central principles, institutions, procedures, and decision-making processes of the political system;
CO 4	To evaluate the basic strengths and weaknesses of the political systems of the Western World;
CO 5	To give knowledge about the theoretical and practical role of state, their governmental organs and the relationship among legislatures, executives and judiciaries.

Course Learning Outcomes (CLOs)

At the end of the course, students will be able to:

CLO 1:	understand the basic concept of political organizations, relations and functions of institutional structure;
CLO 2:	think critically why countries choose different mechanisms and institutions for governing and why these choices are important;
CLO 3:	develop a sound knowledge about the major concepts, and tools of political science and the relationships among organs of the government and their application in different countries.

Course Contents:

Chapter No.	Topics	Teaching Learning Strategy	CLOs
1	Introducing Political Science, Definition, Scope, Nature, State and Society, State and Government,	Lecture Assignment	CLO 1

	Theories of the Origin of State. Fundamental Concepts: Sovereignty, Law, Liberty, Rights, Nation, Nationalism, and Internationalism		
2	Constitution: Definition, Necessity, Methods of Establishing Constitution, Characteristics of a Good Constitution, supremacy of the Constitution, amendment process	Lecture Assignment	CLO 1
3	Forms of Government: Democracy and Dictatorship, Parliamentary and Presidential, and Unitary and Federal	Lecture Assignment	CLO 1 CLO 2
4	Organs of State: Legislature-Its Importance and Functions, Unicameral and Bi-cameral Legislature, Decline of the Power of the Legislature; Executive – Its Importance and Functions, Growing Power of the Executive, Cabinet Dictatorship; Judiciary – Its Importance and Functions, Independence of Judiciary, Judicial Review	Lecture Assignment	CLO 2
5	Separation of Powers, Representation: Proportional and Majoritarian Representation, and Consociationalism	Lecture Assignment	CLO 3
6	Political party: Origin and development of Political parties, Coalition Politics, Functions of Pressure Groups, Difference between Political Parties and Pressure Groups, Bureaucracy and Pressure Groups	Lecture Assignment	CLO 1 CLO 2
7	Public Opinion: Its Meaning, and Significance State and Civil Society, Governance, Good Governance	Lecture Assignment	CLO 3

List of Books

Ball, A. R. & Peters B. G. (2000). *Modern Politics & Government*, New York, Seven Bridges Press.

Islam, M. (2005). *Constitutional law of Bangladesh*, BILIA, Dhaka.

James Q. Wilson. (1995). *Political Organizations*, Princeton, Princeton University Press.

Josep M. C. (2003). *Political Institutions*, Oxford, Oxford University Press.

Laski H. J. (1967). *A Grammar of Politics*. (5th ed.). New York, Harper Collins Publishers Ltd.

Lijphart A. (1977). *Democracy in Plural Societies: A Comparative Exploration*, New Haven, Yale University Press.

Lijphart A. (2008). *Thinking about Democracy: Power Sharing and Majority Rule in Theory and Practice*, Abingdon, Routledge.

March, J.G. & Olsen, J.P. (1989), *Rediscovering Institutions: The Organizational Basis of Politics*, New York, Free Press.

Wilson, J. Q. (1995). *Political organizations*. Princeton University Press.

Course Code:	211709	Marks: 100	Credits: 4	Class Hours: 60
Course Title:	Introduction to Philosophy			

Course Description

This course introduces psychology students to the fundamental questions of philosophy and their relevance to human thought and behavior. It covers key topics such as knowledge, truth, reality, mind–body relation, values, freedom, and the existence of God. The course emphasizes the connection between philosophical reasoning and psychological inquiry.

Through lectures, discussions, and debates, students will learn to analyze arguments. It helps psychology students strengthen critical thinking, logical reasoning, and an understanding of the philosophical foundations of human behavior.

Course Objectives

The objectives of this course encompass:

- Introduce students to the core areas of philosophy (metaphysics, epistemology, ethics, etc.
- Introduce students to the core areas of philosophy (metaphysics, epistemology, ethics).
- Familiarize students with major philosophical figures and their contributions.
- Develop students' abilities to understand, analyze, and evaluate philosophical arguments.
- Enhance students' critical thinking, reasoning, and communication skills.
- Encourage students to develop their own philosophical perspectives and engage in philosophical discussions.

Course Learning Outcomes (CLOs)

After going through the course, students will be able to learn the following outcomes:

CLO1	Students should be able to define and explain key philosophical terms and concepts.
CLO2	Students should be able to identify premises and conclusions in arguments, distinguish different types of arguments, and evaluate their validity and soundness.
CLO3	Students should be able to apply philosophical methods to analyze problems and develop reasoned positions on philosophical issues.

CLO4	Students should be able to formulate their own philosophical perspectives on key issues and support them with reasoned arguments.
CLO5	Students should be able to participate in philosophical discussions, presenting their views clearly and respectfully, and responding to the arguments of others.
CLO6	Students should be able to analyze information, identify biases, evaluate evidence, and form well-reasoned judgments.
CLO7	Students should be able to apply ethical principles to real-world situations and evaluate the ethical implications of different actions and decisions.
CLO8	Students should be able to understand the historical development of philosophical ideas and their relationship to broader social and cultural contexts.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLO/ PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓											
CLO2			✓							✓		
CLO3							✓					
CLO4			✓							✓		
CLO5						✓						✓
CLO6		✓							✓			
CLO7								✓				
CLO8			✓			✓				✓		

Course Contents:

Chapter	Topics	Teaching Learning Strategy	CLOs
1	Origin, nature & scope of philosophy; Philosophy and its related subjects: science, religion, life and common-sense	Lecture, Class Discussion, Reading Assignments	CLO1, CLO8

2	Epistemological problems: rationalism, empiricism, criticism and intuitionism	Lecture, Debate, Group Work	CLO1, CLO2, CLO3
3	Object of knowledge: idealism and realism	Lecture, Analytical Exercises, Case Study	CLO2, CLO3
4	Theories of truth: correspondence theory, coherence theory and pragmatic theory	Lecture, Group Discussion, Critical Analysis	CLO2, CLO6
5	Concept of space, time and matter; Nature of life: mechanism, vitalism and the concept of emergence	Lecture, Question-Answer, Reflective Writing	CLO1, CLO3, CLO6
6	Creation and evolution, theories of evolution: mechanical, teleological, creative and emergent evolution.	Lecture, Multimedia Presentation, Class Debate	CLO3, CLO4, CLO6
7	Theories of reality: monism, dualism and pluralism, idealism and materialism	Lecture, Problem-Solving Tasks, Group Assignment	CLO2, CLO3, CLO4
8	Nature of mind and mind-body relation. Idea of God : proofs for the existence of God	Lecture, Case Study, Group Discussion	CLO2, CLO3, CLO6
9	Nature of value and its classification: intrinsic and extrinsic value Freedom of will and immortality of the soul	Lecture, Debate, Argument Analysis	CLO3, CLO6, CLO7

. **List of Books**

Matin, A. (1968), *An outline of philosophy*. Mullick Brothers, Dhaka.

Patrick, G. T. W. (1935), *An introduction to philosophy*. Houghton Mifflin Company, New York.

Titus, H. H. (1944), *Living issues in philosophy.* American Book Company, New York.

ইসলাম, আনিসুল. (২০০২). *জগৎ জীবন দর্শন* (২য় সংস্করণ), শিক্ষা প্রকাশনী, ঢাকা।

ইসলাম, আনিসুল. (২০০২). *দার্শনিক সমস্যা ও সমকাল*, মৌলী প্রকাশনী, ঢাকা।

রহমান, এম. মতিউর. (২০১৪). *দর্শনের চিরায়ত সমস্যা [প্রথম খণ্ড: জ্ঞানতত্ত্ব]*, অবসর প্রকাশনা সংস্থা, ঢাকা।

রহমান, এম. মতিউর. (২০১৪). *দর্শনের চিরায়ত সমস্যা [দ্বিতীয় খণ্ড: পরাতত্ত্ব]*, অবসর প্রকাশনা সংস্থা, ঢাকা।

রহমান, এম. মতিউর ও ঢালী, মুহাম্মদ আবুল হাই .(২০১০). *দর্শনের মূলনীতি,* নভেল পাবলিশিং হাউজ, ঢাকা।

দেব, গোবিন্দ চন্দ্র .(১৯৭৫). *তত্ত্ববিদ্যাসার*, কেন্দ্রীয় বাঙলা উন্নয়ন বোর্ড, ঢাকা।

বারি, মুহাম্মদ আব্দুর .(১৯৮৮). দর্শনের *কথা*, দিদার পাবলিশিং হাউজ, ঢাকা।

নূরনবী, মোঃ.(১৯৯০). *র্শনের সমস্যাবলী* (৫ম সংস্করণ), লাইব্রেরী, ঢাকা।

আলম, রশীদুল .(১৯৭৯).*দর্শনের ভূমিকা* (২য় সংস্করণ), সাহিত্য কুটির, বগুড়া।

Course Code:	213207	Marks:100	Credits: 4	Class Hours:60
Course Title:	Introduction to Geography and Environment			

Course Description

This course provides the basic concept of geography and earth system science with the definition, meaning, nature and scope. The content of the course is also described the astronomical aspects of the earth. After completion of this course students will learn about the universe, galaxy, solar system and the earth along with the understanding of latitude, longitude and global timing.

Course Learning Outcomes (CLOs)

CLO1	Able to understand the introduction of geography and environment, and their relations giving emphasis on astronomical knowledge.
CLO2	Able to describe the creation of Universe, formation of galaxy, Solar System and differences between Earth and other planets in the Solar System
CLO3	Able to explain earth's co-ordination system, global calendar system and its application in the real life.
CLO4	Able to analyze the Earth's movement (rotation and revolution) and its effects on time, day length, seasons, and illumination.
CLO5	Able to interpret time zone differences and calendar systems from different cultural and geographic perspectives.
CLO6	Able to evaluate the interrelationships between environment, society, and geography with emphasis on human-environment interaction.
CLO7	Able to collaborate in group activities and demonstrate critical thinking through map reading, timeline analysis, and concept mapping.

Mapping CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	✓					✓					
CLO2	✓	✓	✓			✓	✓		✓	✓		
CLO3	✓	✓			✓							
CLO4	✓	✓	✓									
CLO5	✓	✓			✓	✓						
CLO6	✓	✓	✓	✓		✓	✓	✓				✓
CLO7	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Course Contents:

Module No.	Contents	Teaching-Learning Strategies	CLOs
1	**Geography and Environment:** Meaning, fundamental concepts, scope, nature and branches. The relationship between geography and astronomy	Lecture (PPT), short video clips, concept mapping, group discussion, Q&A, think–pair–share	CLO1
2	**Universe:** Big-Bang theory, creation of primary elements, formation of galaxy, and stars.	Multimedia lecture, group activity on cosmic timeline, guided Q&A	CLO2
3	**Solar System:** Planets and Satellites, Sun, Earth and Moon.	PPT-based lecture, discussion, solar system model interpretation, visual quiz	CLO2
4	**Earth System:** Form and shape of the earth including oblate and spheroid curvature of the earth. Common centre of gravity. Polar axis and equatorial axis.	Diagram demonstration, 3D globe activity, interactive lecture, problem-solving	CLO3
5	**Earth's Circumference and Area:** Longitude and Latitude determination. Prime meridian. Length of latitudes and longitudes. Great circle. Rhumb line.	Practical mapping task, group exercise, case examples, peer feedback discussion	CLO3 & CLO7
6	**Rotation and Revolution:** Illumination of the globe. Circle of illumination. Almanac and analema. Length of day and night, and altitude of the sun. Solstices and equinoxes. Twilight - types, determination.	Lecture with animation, diagram-based demonstration, almanac analysis, class discussion	CLO4 & CLO7

7	**Time:** Longitude and time. Local and standard time, division of the earth according to standard time. Standard time of USA, India and Bangladesh. International Date Line. Calendar-Astronomical, Jewish, Mohammedan, Roman, Julian, Gregorian and World. **Time Zone Calculation:** Mathematics of Time Zone Calculation, International and Local Time Zones.	Timeline construction, comparative charting, calendar workshop, case-based learning	CLO5 & CLO7
8	**Environment:** Meaning, structure, types and components of Environment; Geography and Environment; Environment and Society, Ecology, Sustainability.	Interactive lecture with PPT, case study analysis on human-environment interaction	CLO6

References

Adhikari, S. (2011). *Fundamentals of geographical thought*. Chaitanya Publishing House, Allahabad, India.

Arny, T., & Schneider, S. E. (1994). *Explorations: An introduction to astronomy*. Mosby.

Berry, B. J. L. (1973). *A paradigm for modern geography*. In R. J. Chorley (ed.), *Directions in geography*. Methuen, London.

Broek, J. O. (1965). *Geography: Its scope and spirit*. Charles E. Merrill, Columbus, Ohio. Branley, F. M. (1975). *Astronomy.* Thomas Y. Crowell & Co., New York.

Singh, S. (2001). *Environmental geography*. Prayag Pustak Bhawan, India.

Strahler, A. H., & Strahler, A. N. (2006). *Introducing physical geography*. Wiley.

Course Code	213107	Marks: 100	Credits: 4	Class Hours: 60
Title	Zoology –I			

Course Objectives

This course is an introduction to basic concepts in biology through study of the major lineages of invertebrate with emphasis on the ontogeny, structure, and function of organ systems. The students will be able to understand the living organisms with a focus on the animal kingdom and their direct impact on our economy.

Course Learning Outcome (CLO): After going through the course, students will be able to learn the following outcomes:

CLO1	**Fundamental Knowledge:** Explain the scope, levels of biological organization, and foundational zoological concepts with emphasis on animal diversity and phylogeny.
CLO2	**Biodiversity & Taxonomy:** Identify and classify major animal phyla and economically significant species using morphological, ecological, and systematic criteria, with focus on Bangladesh's fauna.
CLO3	**Practical Research Skills:** Apply field and laboratory techniques to study animal anatomy, life cycles, and biodiversity, including calculations of diversity indices for local ecosystems.
CLO4	**Applied Zoology:** Analyze the role of zoology in agriculture, fisheries, and public health, proposing context-specific solutions for Bangladesh.
CLO5	**Ethics & Communication:** Present scientific findings on animal systems through structured reports or presentations, adhering to ethical guidelines in wildlife research and conservation.

Mapping of CLOs with PLOs

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10
CLO1	√									
CLO2		√			√					
CLO3			√	√						
CLO4									√	√
CLO5							√	√		

Course Contents:

Topics	Teaching Learning Strategy	CLOs
Group-A: General Zoology		
1. **Introduction to Zoology**: Definition and scope of zoology. Foundation of animal life: Level of organization (protoplasmic, cellular, tissue, organ, organ system, organism, species, individual, population, community, fauna, biota, ecosystem, biosphere, biodiversity. Cells: Cell and cell theory, structure and functions of cell organelles. Gametogenesis: Spermatogenesis and oogenesis; placentation. Classification of animals: Animal kingdoms; classification up to phyla based on organization, symmetry, coelom and phylogeny; different taxa, Linnean hierarchy and nomenclature	Lecture, and Group Discussion,	CLO1
2. Biodiversity: Concept, types, components, importance and values, bioresources – threats and conservation importance, estimation and calculation process of biodiversity, faunal diversity in Bangladesh	Lecture, Group Discussion, Assignments and tutorial sessions	CLO2 CLO3
3. Methods of Studying Animals: Survey of aquatic and terrestrial animals; Methods of animal collection, transportation, preservation, identification, description, tagging	Lecture, Group Discussion, and Assignments	CLO3 CLO5
4. **Systematics**: Identifying characteristics, life cycle, habitats, and economic importance of **a. Zooplankton:** Radiolarians, foraminiferans, and dinoflagellates, cnidarians, and crustaceans b. **Parasitic and vector insects:** Fleas, lice, mites, ticks, mosquitoes, and various flies c. **Aquatic Insects**: Trichoptera, Ephemeroptera, True bugs, Odonata, Plecoptera, Beetles, Caddisflies, Non-biting midges, Coleoptera, Diptera, Dragonflies and damselflies, Mayflies, Stoneflies	Lecture, Group Discussion, Lab demo and Field visit	CLO2
Group B: **Type Study**		
1. **Short study:** Diagnostic characteristics and economic importance of the following non-chordates with examples – Apicomplexa, Ciliophor, Porifera, Cnidaria, Ctenophora, Platyhelminthes, Gastrotricha, Nematomorpha, Rotifera, *Trypanosoma, Leishmania, Amoeba, Monocystis, Nosema, Spongilla, Adamsia, Fasciola, Schistosoma, Convoluta, Diphyllobothrium, Ancylostoma, Enterobius, Wuchereria, Loa, Trichuris, Loligo, Octopus, Mytillus, Sepia, Tubifex, Limulus, Eupagurus, Squilla*, moths & butterflies, termite,	Lecture, Group Discussion, and Assignments	CLO2

bugs and bees		
2. **Descriptive study:** Systematic position, habitats, external morphology, organ systems such as digestion, movement, circulation, respiration, excretion, nervous, and reproduction and development; food and feeding habits– *a.* Phylum Sarcomastigophora: *Euglena* *b.* Phylum Ciliophora: *Paramecium* c. Phylum Porifera: *Scypha* d. Phylum Cnidaria: *Obelia* e. Phylum Nematoda: *Ascaris* *f.* Phylum Platyhelminthes: *Taenia* *g.* Phylum Mollusca: *Pila* h. Phylum Arthropoda: *Prawn* i. Phylum Echinodermata: *Astropecten* *j.* Phylum *Hemichordata: Balanoglossus*	Lecture, Group Discussion, and tutorial sessions	CLO1
3. Applied Zoology: a. Introduction to the major fields of Applied Zoology: Entomology, Fisheries, Wildlife Biology, Parasitology and others. b. Agricultural Pests: Major Pests of Rice, Jute, Sugarcane & stored grain. c. Integrated Fish Farming: Types; Poultry, Livestock and Paddy-cum -Fish Culture. d. Poultry farming: g of poultry farming, diseases of poultry and their control, economic importance of poultry and their impacts on socio-economic condition of Bangladesh.	Lecture, Group Discussion, Assignments and Field visit	CLO4 CLO5

List of Books:

Campbell, N., & Reece, J. (2005). *Biology*, (7th ed.). 2005, Pearson

Gaston, K. J., & Spicer, J. I. (2013). *Biodiversity: An Introduction (2nd ed.)*. John Wiley and Sons Inc., New York.

Hickman, C.P., Keen, S. L., Eisenhour, D.J., Larson. A. & I'Anson, H. (2023). *Integrated Principles of Zoology* (19th ed.). McGraw-Hill Co. Inc., New York, USA.

Nigam, H. C. (2013): *Biology of Non-chordates.* Vishal Publishing Co., India

Parker, T. J. & Haswell, W. A. (2005): *Text Book of Zoology,* Vol. I, Macmillan

Ruppert, E. E., Fox, R. S., & Barnes, R. D. (2006). *Invertebrate Zoology* (8th ed.). Holt Saunders International edition

Singla, M., Kaur, A., Sitre, S. R. & Gupta, A. (2023). Applied Zoology and Animal Biotechnology. (1 st ed.). AG Publishing House, India.